

SOLUTION REQUIREMENTS

EDUGENIE – AI-POWERED LEARNING ASSISTANT

Team Members:

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FUNCTIONAL REQUIREMENTS

FR-01 – Question Answering

The system shall accept a natural-language question and route it to the question-answering module.

FR-02 – Concept Explanation

The system shall accept a topic and generate a simplified explanation using the configured local explanation model.

FR-03 – Quiz Generation

The system shall accept supplied learning material or a topic and generate a multiple-choice quiz.

FR-04 – Text Summarization

The system shall accept longer educational text and return a concise summary containing important information.

FR-05 – Learning Path

The system shall accept a learning topic and generate a structured progression from basic to advanced topics.

FR-06 – Web Interface

The system shall provide controls through an HTML/CSS web interface for the major learning functions.

FR-07 – Backend Routing

The FastAPI backend shall receive requests, invoke the appropriate module, and return generated output.

FR-08 – AI Integration

The system shall integrate Gemini through the Gemini API for cloud-based generation tasks.

NON-FUNCTIONAL REQUIREMENTS

NFR-01 Usability

The interface should make the five learning capabilities straightforward to access.

NFR-02 Maintainability

Learning functions should remain separated into individual modules.

NFR-03 Security

API credentials should be supplied through environment configuration and not exposed in public source code.

NFR-04 Availability

The local FastAPI application should be runnable in a Python environment with the required dependencies.

NFR-05 Performance

The application should provide results after the relevant AI operation completes. External Gemini response time may depend on network and service conditions.

NFR-06 Extensibility

The modular structure should allow additional learning features to be added later.